Notes: Greenwood, Seshadri, and Yorukoglu (ReStud, 2005)

Engines of Liberation

Contents

1 Big picture

- **Question.** Can the household consumer-durable revolution—electricity, washing machines, refrigerators, dryers, and other labor-saving home technologies—help explain the long-run rise in married female labor-force participation in the United States?
- **Setting.** The paper studies the period roughly from 1900 to 1980 and embeds Beckerian household production inside a dynamic general equilibrium model with adoption of labor-saving household durables.
- **Core challenge.** A standard growth model in which market wages simply rise over time does *not* generate a rise in female labor-force participation. Something non-neutral has to change inside the household production technology.
- **Main mechanism.** New household durables substitute for housework. Because they save time at home, they reduce the shadow cost of market work for women. Richer households adopt earlier because the new technologies are initially expensive and then become cheaper over time.
- **Why this matters.** The paper gives a structural explanation for a central transformation of the twentieth century: the movement of married women from home production into market work.
- **Main quantitative result.** In the baseline model with lumpy durables and indivisible female labor supply, technological progress in the household sector accounts for about **28 percentage points** of the model's **51 percentage point** rise in female labor-force participation. The narrowing of the gender gap alone accounts for only about **10 percentage points**.
- **Interaction effect.** Household durables do not just directly raise labor-force participation. They also make female labor supply more responsive to wages, so the decline in the burden of housework and the narrowing of the gender gap reinforce each other.
- **Robustness result.** In the alternative model with divisible effort and divisible durables, the durable goods revolution still explains about **half** of the rise in female labor-force participation between 1900 and 1980.
- **Economic takeaway.** The paper's central claim is that household technology is not a side issue. It is a major part of the explanation for the rise in married women's market work.

2 Data and measurement (key objects)

2.1 Historical facts and motivating evidence

- The paper is motivated by long-run historical series on the diffusion of **electricity, central heating, running water, flush toilets, refrigerators, washers, dryers, dishwashers, microwaves,** and other appliances.
- The share of households with these basic facilities and appliances rises dramatically over the twentieth century. The diffusion curves in Figure 1 show the broad household-technology transition that motivates the model.
- Investment in household appliances as a share of GDP **more than tripled** over the century. By 1988 it was about **0.5% of GDP** and about **2.9% of total investment spending**.

- The stock of appliances as a share of GDP also roughly **doubled** over time.
- Richer households tended to adopt appliances earlier than poorer households, consistent with the paper's adoption mechanism.

2.2 Time use, labor-market facts, and the gender gap

- In 1900 the average household spent about **58 hours per week** on housework; by 1975 this had fallen to about **18 hours per week**.
- Over the same period, the number of paid domestic workers declined sharply, consistent with housework becoming less labor intensive.
- Female labor-force participation rose steadily from about **5% in 1900** to about **50% in 1980**.
- Women's earnings relative to men's rose only modestly over the same broad period: from about **0.48–0.50** early in the century to about **0.59–0.60** later on.
- This combination of facts is important: female labor supply rose enormously, but the gender wage ratio did not move enough on its own to make that rise obvious in a standard model.

2.3 Households, technology, and heterogeneity in the model

- A household consists of a male and a female.
- Males always work a fixed market workweek.
- Females choose whether to work in the market or stay at home; in the baseline model this decision is **indivisible**.
- Households differ by an ability or income type λ, drawn from a lognormal distribution.
- Household production combines **durables** and **housework**. In the baseline model, durables are **lumpy** and housework is effectively tied to the participation margin.
- The model distinguishes an **old home technology** and a **new labor-saving technology**. Adopting the new technology is costly when prices are high but increasingly attractive as the price of durables falls.

2.4 Calibration targets and key quantitative objects

- The model period is **5 years**.
- In the market sector, labor's share is set to **70%**, and the annual depreciation rate of business capital is set to **10%**.
- A full-time workweek corresponds to **40 hours out of 112 nonsleeping hours**, so $\omega = 0.36$.
- The household production calibration uses the housework facts to set $\eta = 0.16$ and $\rho\eta = 0.52$.
- The stock of appliances rises from **\$66 per capita in 1925** to **\$528 in 1980** (in 1982 dollars), implying $\kappa = 8$ in the baseline household-technology comparison.

- For the baseline calibration, the paper sets $J = 10$ (a 50-year working life), $\beta = 0.965$, and the standard deviation of log ability to $\sigma = 0.70$.
- In the 1980 steady state, the paper chooses $\mu = 0.33$, $\nu = 0.20$, and $q = 0.03$ to match female labor-force participation near **51%** and appliance investment near **0.45% of GDP**.

3 Models and empirical specifications (all equations + interpretation)

3.1 Preferences

For an age-j household, lifetime utility is

$$\sum_{i=j}^{J} \beta^{i-j} \left[\mu \ln m_i + \nu \ln n_i + (1-\mu-\nu) \ln l_i\right],$$

where:

- m_i is market consumption,
- n_i is home-produced consumption,
- l_i is leisure.

Interpretation. Households value market goods, home goods, and leisure. This is the Beckerian structure that makes time allocation and household production central to the analysis.

3.2 Household production and the durable-goods revolution

In the baseline model, household production is Leontief:

$$n = \min\{d, \zeta h\},$$

where:

- d is the stock of household durables,
- h is housework,
- ζ is labor-augmenting household technology.

The old technology is parameterized as

$$d = \delta, \qquad h = \rho\eta, \qquad \zeta = \frac{\delta}{\rho\eta},$$

while the new technology is

$$d' = \kappa\delta, \qquad h' = \eta, \qquad \zeta' = \frac{\kappa\delta}{\eta} = \kappa\rho\zeta.$$

Interpretation.

- The new technology uses more effective household capital services.
- It requires less housework to generate home output.
- The key economic meaning of the durable goods revolution is therefore **labor saving at home**. This is exactly what frees women for market work.

3.3 Market production and resource constraints

Market output is produced by a standard neoclassical technology:

$$y = \xi k^{\alpha}(zl)^{1-\alpha},$$

where k is business capital, l is aggregate labor, and z is labor-augmenting market productivity.

Output is allocated to market consumption, business investment, and household capital:

$$m + i + d = y.$$

Business capital evolves according to

$$k' = \chi k + i.$$

Interpretation. The household durable stock is treated as part of the economy's resource allocation problem. The home technology is not "free"; it competes with market uses of output.

3.4 Household dynamic programming problem

Let $V_j(a, \tau, \lambda)$ denote the value of an age-j household with assets a, technology state $\tau \in \{0, 1, 2\}$, and ability λ.

When the household has **not** adopted the new technology in the current period ($\tau = 0$),

$$\begin{aligned} V_j(a, 0, \lambda) = \max \Big\{ &\max_{a'} \big[\mu \ln(w\lambda\omega + \phi w\lambda\omega + ra - a') + \nu \ln \delta \\ &\quad + (1 - \mu - \nu) \ln(2 - 2\omega - \rho\eta) + \beta \max\{V_{j+1}(a', 0, \lambda), V_{j+1}(a', 1, \lambda)\}\big], \\ &\max_{a'} \big[\mu \ln(w\lambda\omega + ra - a') + \nu \ln \delta \\ &\quad + (1 - \mu - \nu) \ln(2 - \omega - \rho\eta) + \beta \max\{V_{j+1}(a', 0, \lambda), V_{j+1}(a', 1, \lambda)\}\big] \Big\}. \end{aligned}$$

When it **adopts** in the current period ($\tau = 1$),

$$\begin{aligned} V_j(a, 1, \lambda) = \max \Big\{ &\max_{a'} \big[\mu \ln(w\lambda\omega + \phi w\lambda\omega + ra - a' - wq) + \nu \ln(\kappa\delta) \\ &\quad + (1 - \mu - \nu) \ln(2 - 2\omega - \eta) + \beta V_{j+1}(a', 2, \lambda)\big], \\ &\max_{a'} \big[\mu \ln(w\lambda\omega + ra - a' - wq) + \nu \ln(\kappa\delta) \\ &\quad + (1 - \mu - \nu) \ln(2 - \omega - \eta) + \beta V_{j+1}(a', 2, \lambda)\big] \Big\}. \end{aligned}$$

If it has already adopted ($\tau = 2$),

$$V_j(a,2,\lambda) = \max\Big\{ \max_{a'} \big[\mu \ln(w\lambda\omega + \phi w\lambda\omega + ra - a') + \nu \ln(\kappa\delta) + (1-\mu-\nu)\ln(2 - 2\omega - \eta) + \beta V_{j+1}(a',2,\lambda)\big], \max_{a'} \big[\mu \ln(w\lambda\omega + ra - a') + \nu \ln(\kappa\delta) + (1-\mu-\nu)\ln(2 - \omega - \eta) + \beta V_{j+1}(a',2,\lambda)\big]\Big\}.$$

The adoption decision is

$$\max_{\tau\in\{0,1\}} V_j(a,\tau,\lambda),$$

so the adoption rule is

$$T^j(a,\lambda) = \begin{cases} 1 & \text{if } V_j(a,1,\lambda) > V_j(a,0,\lambda), \\ 0 & \text{if } V_j(a,1,\lambda) \le V_j(a,0,\lambda). \end{cases}$$

Interpretation.

- The female participation decision is discrete: work or stay home.
- The adoption decision is also discrete: buy the new technology now or wait.
- The model therefore captures both the **extensive margin of labor supply** and the **extensive margin of technology diffusion**.

3.5 Competitive equilibrium and balanced growth

The labor market clears when male and female labor supplies add up to aggregate labor demand,

$$l = J\omega \int \lambda\, L(d\lambda) + \phi\omega \sum_{j=1}^{J} \int \lambda P^j(\lambda)\, L(d\lambda).$$

Aggregate business capital satisfies

$$k' = \sum_{j=1}^{J} \int A^j(\lambda)\, L(d\lambda).$$

Factor prices equal marginal products:

$$w = (1-\alpha) z\xi (zl/k)^{-\alpha},$$

$$r' = \alpha\xi (z'l'/k')^{1-\alpha} + \chi.$$

A key lemma shows that along a balanced-growth path,

$$A^j_{t+1}(\lambda) = \gamma A^j_t(\lambda), \qquad P^j_{t+1}(\lambda) = P^j_t(\lambda), \qquad T^j_{t+1}(\lambda) = T^j_t(\lambda).$$

Interpretation. The adoption and participation rules are stationary in growth-adjusted terms. This makes the model tractable and lets the authors compare steady states and transitions.

3.6 Why the standard home-production model fails

A crucial remark in the paper is that if one removes the non-neutral change in household technology, then growth in market wages alone does *not* increase female labor-force participation.

Interpretation. In a standard balanced-growth home-production model, the rise in wages generates offsetting income and substitution effects. So the paper needs a **technological change inside the household sector**, not just general economic growth.

3.7 Partial-equilibrium mechanism: falling durable prices and adoption

In a partial-equilibrium experiment, the paper studies a lifetime with a fixed interest rate and a durable price path that falls at about **8.3% per year**.

Main theoretical implications:

- Nobody adopts immediately when prices are very high.
- Wealthier households adopt earlier.
- Adoption is associated with a jump up in female labor-force participation.
- Along a balanced-growth path, the date of adoption is non-increasing in type when periods are sufficiently short.

Interpretation. The model generates gradual diffusion because households face a timing problem: buy an expensive labor-saving technology now, or wait for it to become cheaper.

3.8 Robustness model with divisible effort and durables

To show the mechanism does not depend on lumpy durables and indivisible labor, the paper studies an alternative representative-household setup with infinite horizon preferences

$$\sum_{j=1}^{\infty} \beta^{j-1} \left[\mu \ln c_j + \nu \ln n_j + (1 - \mu - \nu) \ln l_j \right],$$

and CES household production

$$n_j = \left[\theta d_j^{\vartheta} + (1 - \theta) h_j^{\vartheta} \right]^{1/\vartheta}, \qquad \vartheta < 1.$$

The household solves

$$\max_{\{c_j, h_j, d_j, l_j\}_{j=1}^{\infty}} \sum_{j=1}^{\infty} \beta^{j-1} \left[\mu \ln c_j + \nu \ln n_j + (1 - \mu - \nu) \ln l_j \right]$$

subject to the intertemporal budget constraint

$$\sum_{j=1}^{\infty} p_j c_j = a_0 + \sum_{j=1}^{\infty} p_j w_j \omega + \sum_{j=1}^{\infty} p_j w_j \left[\phi_j (1 - h_j - l_j) - q_j d_j \right].$$

The optimality conditions imply

$$h_j = \nu (1 - \beta) \beta^{j-1} \frac{a_0 + \sum_i p_i w_i (\omega + \phi_i)}{p_j w_j (\phi_j + q_j Q_j)},$$

$$l_j = (1-\mu-\nu)(1-\beta)\beta^{j-1}\frac{a_0 + \sum_i p_i w_i(\omega+\phi_i)}{p_j w_j \phi_j},$$

where

$$Q_j \equiv \left(\frac{1-\theta}{\theta}\frac{q_j}{\phi_j}\right)^{1/(\vartheta-1)}.$$

The paper then proves two comparative-static results:

- if durables and housework are Edgeworth–Pareto substitutes ($\vartheta > 0$), a fall in the rental price of durables lowers housework and raises market work;
- if durables and housework are substitutes, a reduction in the gender gap also lowers housework and raises market work.

Interpretation. The deeper force is substitution between durables and housework. The lumpy/adoption structure is useful for matching diffusion, but the economic logic is more general.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- A standard balanced-growth home-production model cannot explain the rise in female labor-force participation because rising wages alone do not shift market work for women.
- A pure wage-story also struggles in the data, because the female-to-male earnings ratio rises only modestly while female labor-force participation rises dramatically.

4.2 What disciplines the model

The paper is not identified by instruments in the modern econometric sense. Instead, it is disciplined by a set of historical targets and model restrictions:

- female labor-force participation around 1980,
- the 1980 female wage ratio,
- appliance investment as a share of GDP,
- time-use facts on housework,
- appliance stock growth,
- the time path of appliance prices,
- the time path of the gender gap.

Interpretation. The credibility of the quantitative exercise comes from matching broad historical moments and then asking whether the calibrated model can reproduce the transition in female labor-force participation.

4.3 Why the adoption margin is important

- Lumpy durables are essential for generating gradual diffusion.
- Indivisible female labor is useful because the historical series is fundamentally an **extensive-margin** participation series.
- The model's cross-household heterogeneity in λ is what generates earlier adoption by richer households.

4.4 Empirical credibility and limitations

- The model uses **perfect foresight** for the future paths of prices and the gender gap; the authors explicitly call this heroic.
- The baseline assumes no second-hand market for durables.
- Other possible forces behind female labor-force participation—such as job composition changes or social norms—are intentionally abstracted from rather than denied.
- In the alternative divisible-labor model, the implied female labor-supply elasticity may be too high and the model predicts too much appliance investment in response to falling prices.

4.5 Relation to the literature

- The paper builds on Beckerian household production.
- It differs from Benhabib, Rogerson, and Wright by focusing on long-run household technological change rather than business cycles.
- It complements Galor and Weil, who emphasize technological change in the *market* sector, by emphasizing technological change in the *household* sector.

5 Tables and figures

5.1 Figure 1 and Figure 2: the household technology revolution

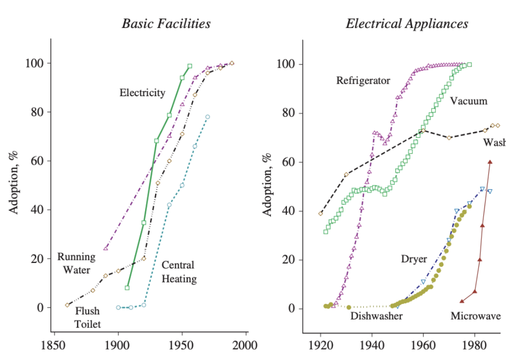

FIGURE 1
The diffusion of basic facilities and electrical appliances through the U.S. economy

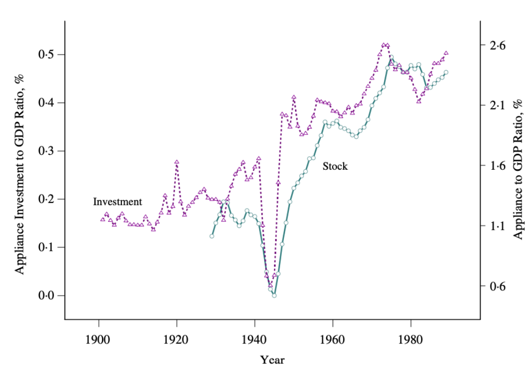

FIGURE 2
Household appliances

Read:

- Figure 1 is the broad historical motivation. It shows the spread of electricity and household appliances through American households.
- Figure 2 shows that appliance investment and the stock of appliances became increasingly important relative to GDP.
- These figures tell you the paper is not about a small margin. It is about a major transformation in household production technology.

5.2 Figure 3: housework and market work

Read:

- Figure 3 is the key reduced-form fact: housework falls sharply while female market work rises strongly.
- The decline in domestic servants reinforces the interpretation that home production itself became less labor intensive.
- The modest movement in the female earnings ratio compared with the huge rise in participation motivates the paper's skepticism about a pure wage-gap explanation.

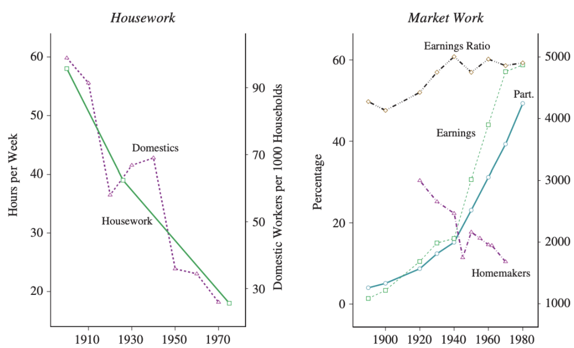

FIGURE 3
Housework and market work

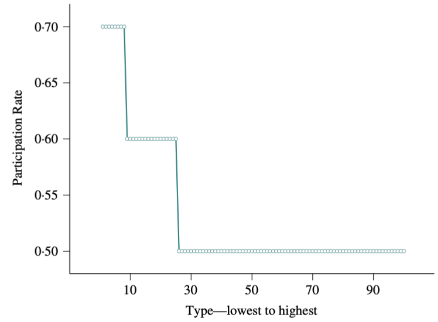

FIGURE 4
Female labour-force participation

5.3 Figure 4, Figure 5, and Figure 6: type heterogeneity, welfare, and adoption timing

Read:

- Figure 4 shows that when the new technology is present, female participation becomes a decreasing function of household type. The fixed cost of durables matters less for richer households.
- Figure 5 gives the welfare interpretation of the household revolution. The welfare gains are so large that a poor family in 1980 can be as well off as a very rich family in 1900.
- Figure 6 shows the mechanism at the individual level: as prices fall, adoption occurs, and adoption immediately raises participation.

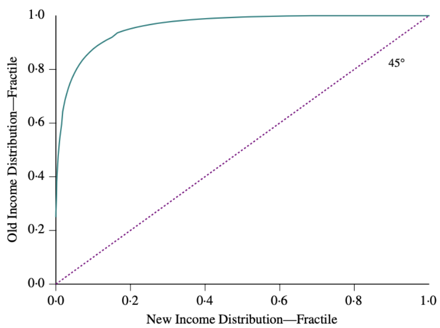

FIGURE 5
An utilitarian view of economic development

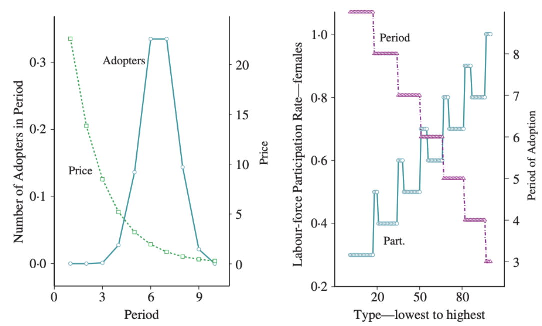

FIGURE 6
The effect of prices on adoption and participation

5.4 Figure 7, Figure 8, Figure 9, Table 1, Figure 10, and Figure 11: transition, diffusion, and welfare

Read:

- Figure 7 provides the crucial exogenous force in the simulation: time prices of appliances fall rapidly, with an index dropping about **10% per year** and the simulation using **8.3% per year** on average.
- Figure 8 shows the main transition path. The model generates a strong rise in female labor-force participation as prices fall and the gender gap narrows.
- Figure 9 decomposes the forces. With no durable goods revolution, participation in 1980 is only about **10%**. With fixed gender gap but falling durable prices, it reaches about **40%**. In the baseline, it reaches about **54%** in 1980 and converges to a steady state near **51%**.
- Table 1 shows the same interaction in steady-state form: for any given gender gap, female participation is much higher when modern household durables are available.
- Figure 10 shows diffusion by income type. Rich households adopt early; poor households adopt late. The aggregate diffusion curve is slow even though eventual adoption is broad.
- Figure 11 shows that welfare gains exceed GDP gains. The paper reports a **30%** rise in GDP in continuously compounded terms but a **173%** gain in welfare measured as compensating variation in market consumption. The durable goods revolution plus a narrower gender gap raise market consumption by **28%**, nonmarket consumption by **208%**, and leisure by **14%**.
- The right panel of Figure 11 implies that the rise in female labor-force participation can account for about **19%** of per-capita real GDP growth between 1900 and 1980 in the model's accounting exercise.

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5.5 Figure 12: robustness with divisible labor and durables

Read:

- Figure 12 shows that the mechanism survives once labor and durables are made divisible.

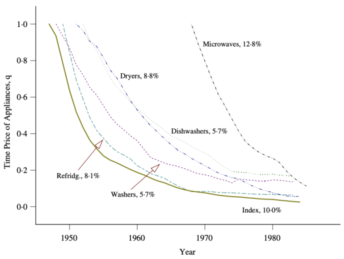

FIGURE 7
Time prices for appliances

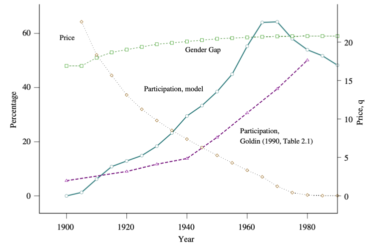

FIGURE 8
Transitional dynamics—the rise in female labour-force participation

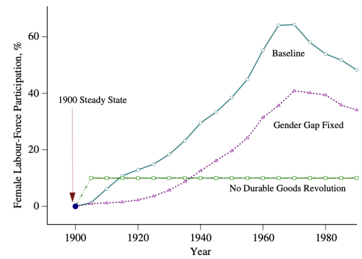

FIGURE 9
Durable goods revolution and gender gap experiments

TABLE 1

Female wages and steady-state labour-force participation

Gender gap, ϕ	Female labour-force participation	
	Without durable goods revol.	With durable goods revol.
0·45	0·00	0·22
0·50	0·00	0·31
0·55	0·10	0·42
0·60	0·10	0·52
0·65	0·20	0·61

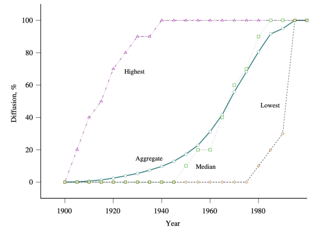

FIGURE 10
Diffusion by type of household

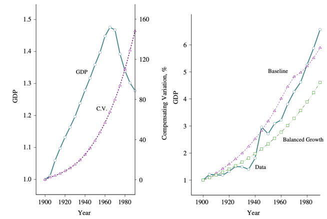

FIGURE 11
The gain in GDP and welfare

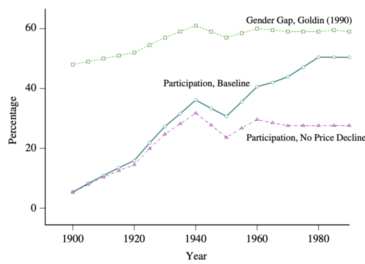

FIGURE 12
The rise in female labour-force participation with divisible effort and non-durable household products and services

- Holding the gender gap path fixed, removing the durable price decline substantially lowers the rise in female labor-force participation.
- The paper concludes from this exercise that labor-saving goods account for about **half** of the rise in female labor-force participation between 1900 and 1980.

6 Short summary

- **Method.** The paper embeds household production and durable-adoption decisions into a dynamic general equilibrium model with heterogeneous households.
- **Key mechanism.** Labor-saving household durables substitute for housework and therefore make market work by women more attractive.
- **Quantitative result.** In the baseline, the household durable revolution explains about **28 percentage points** of the model's **51 percentage point** rise in female labor-force participation; in the alternative divisible model, it explains about **half** of the observed rise.
- **Takeaway.** The paper argues that household technological progress is a first-order explanation for the long-run rise in married women's market work.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that technological progress in the *household sector* can quantitatively explain a major part of the rise in female labor-force participation.
- It also shows that the diffusion of household technology interacts strongly with the narrowing of the gender gap.
- Household durables are therefore not a minor background force; they are central to long-run labor-supply change.

7.2 Economic intuition

- Market work by women is costly when housework is time intensive.
- Labor-saving durables reduce the home-time burden of market participation.
- As appliance prices fall, more households adopt, and that adoption raises female market work.
- The reason the wage effect becomes stronger after adoption is that women are no longer tied as tightly to home production.

7.3 Takeaway

This paper is important because it changes the way to think about twentieth-century labor supply. Instead of treating female labor-force participation as driven mainly by market wages or social change, it places the household production technology at center stage. The core message is simple and powerful: when the home becomes more capital intensive and less labor intensive, women are freed to work in the market.